2.10.1 Newton's law of gravitation

Two masses attract with a force proportional to their mass product and inversely proportional to the square of their centre-to-centre separation. Their forces are equal and opposite. The simple expression applies to point masses or non-overlapping spherically symmetric bodies. Use centre separation, not surface separation or altitude alone.

$$F = \frac{GMm}{r^2}$$

F: attractive force magnitude (N); M, m: masses (kg); r: centre separation (m); G = 6.67 × 10^{-11} N m^2 kg^{-2}. Point masses or non-overlapping spherically symmetric bodies.

2.10.2 Gravitational field strength

Gravitational field strength is force per unit test mass. It points in the direction of gravitational force and is numerically equal to local free-fall acceleration. Outside a spherical body, it decreases with the square of distance from the centre. It is independent of the test mass. At altitude, the distance from the centre is the body's radius plus the height above its surface.

$$g = \frac{GM}{r^2}$$

$$\vec{g} = \frac{\vec{F}}{m_0}$$

g: gravitational field strength (N kg^{-1}), numerically free-fall acceleration (m s^{-2}); M: source mass (kg); r: distance from its centre (m). Outside a spherical mass; field points inward.

g: gravitational field strength (N kg^{-1}); F: force on test mass (N); m_0: test mass (kg).

2.10.3 Speed in a circular orbit

In an unpowered circular orbit, gravity supplies the satellite's centripetal resultant force. Equating the two expressions gives orbital speed and angular speed. Around the same central body, a larger radius gives lower speed and angular speed but a longer period. Measure radius from the body's centre.

A geostationary satellite follows a circular orbit in Earth's equatorial plane, in the same direction as Earth's rotation, with the same rotation period of about 24 hours. It therefore stays above the same ground position. Orbital radius is Earth's radius plus the satellite's altitude.

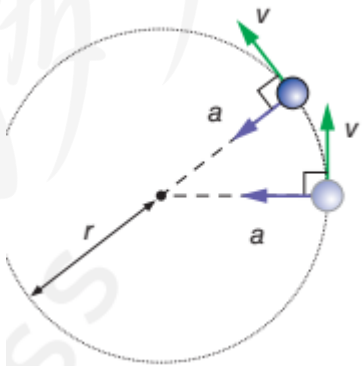
$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$$

$$v = \sqrt{\frac{GM}{r}}$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

Circular orbit: gravity supplies the centripetal force. M: central mass; m: orbiting mass; r: centre-to-centre distance; v: orbital speed; ω: angular speed. M is treated as stationary.

Circular orbit around central mass M (kg); orbital radius r (m) is measured from its centre, not the surface. G = 6.67 × 10^{-11} N m^2 kg^{-2}; v: m s^{-1}; ω: rad s^{-1}.



Velocity is tangential to the circle, while centripetal acceleration points towards its centre.